

ATNA

Air Traffic Network Analysis

Interface walkthrough

Structural criticality and scenario modelling for the U.S. domestic flight network. Every panel is a live capture of the running application.

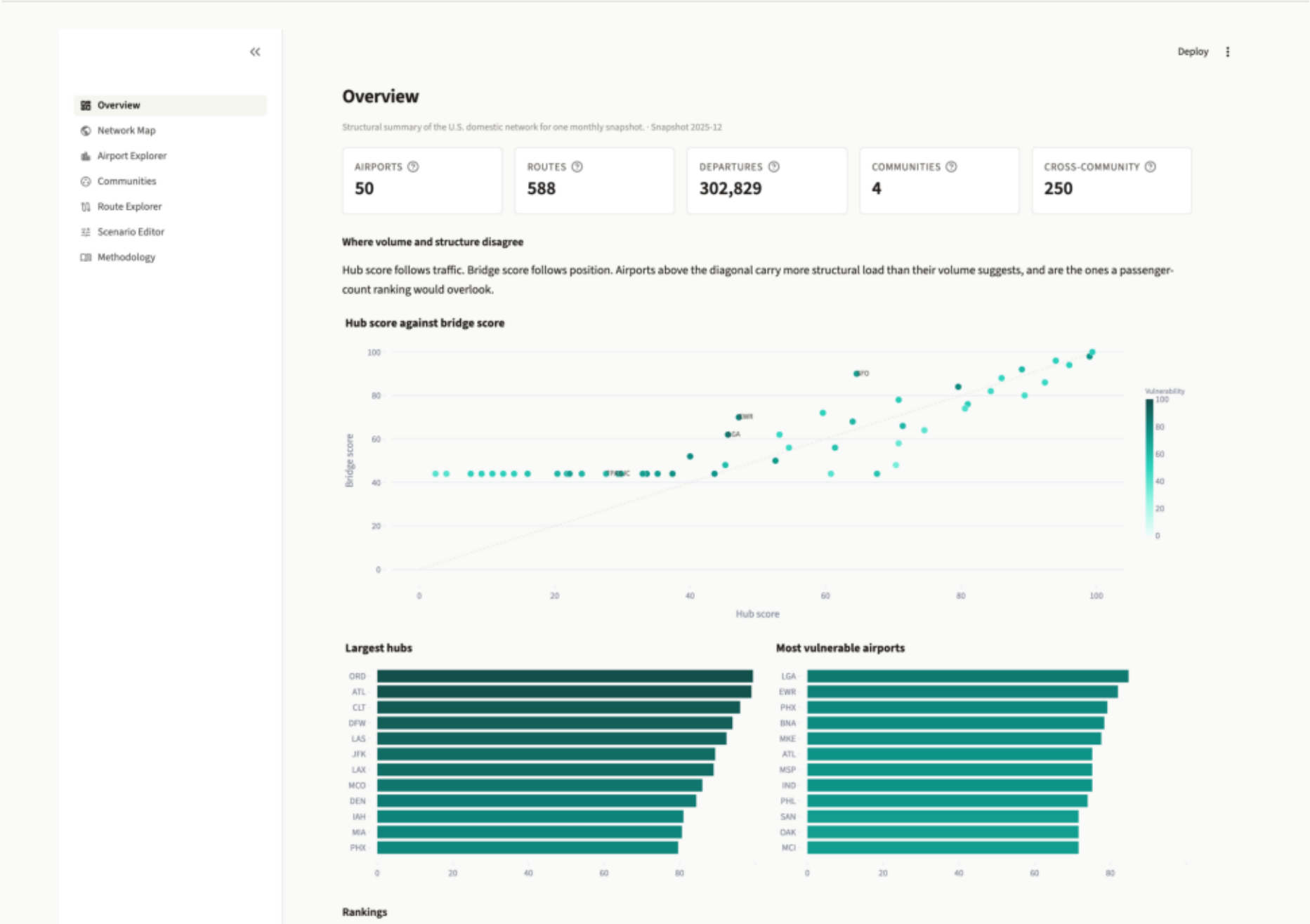
Snapshot 2025-12

Pages 8

Generated 2026-08-19

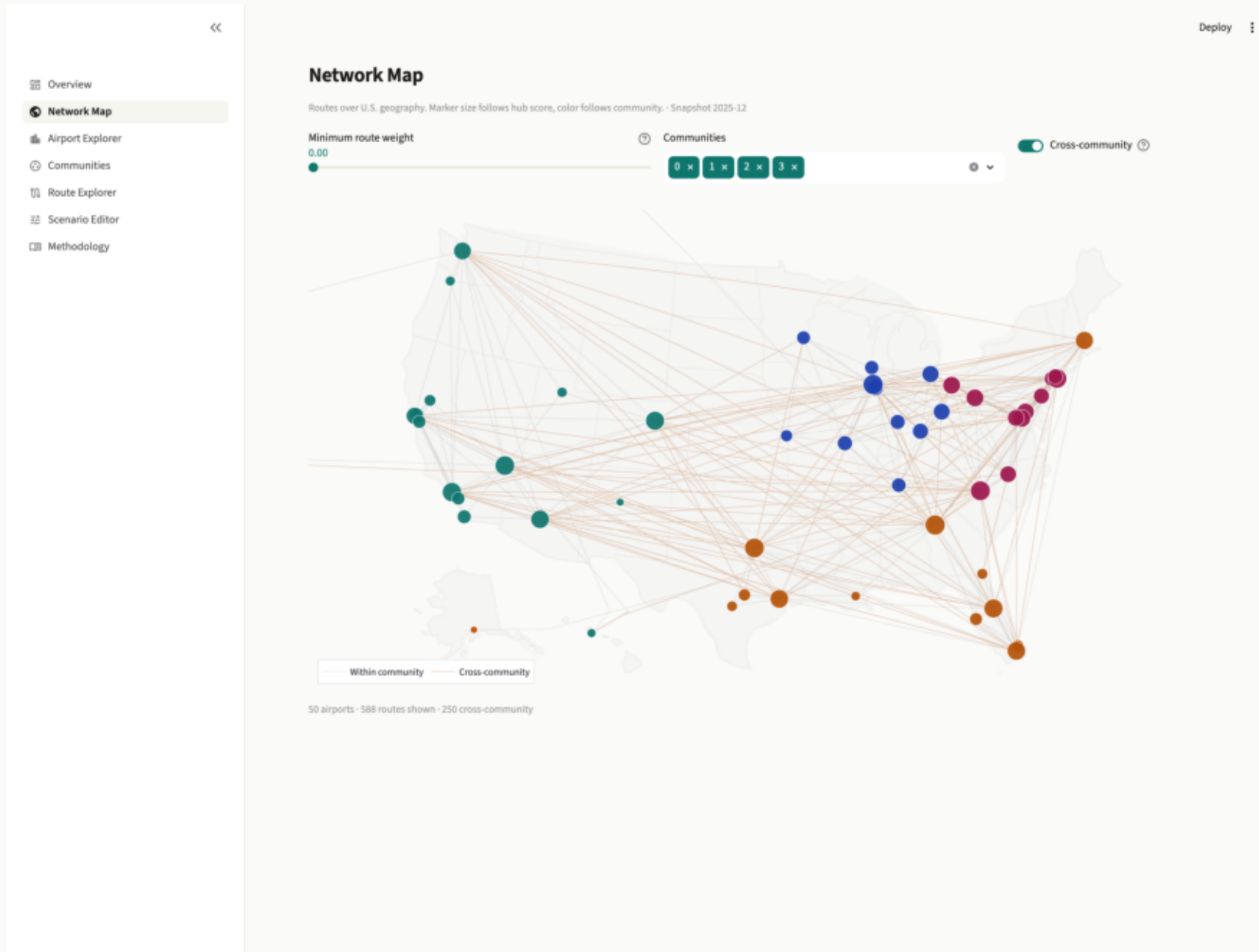
Overview

Snapshot totals, then the projects central claim: hub score against bridge score with the diagonal drawn in. Airports above the line carry more structural load than their traffic implies.



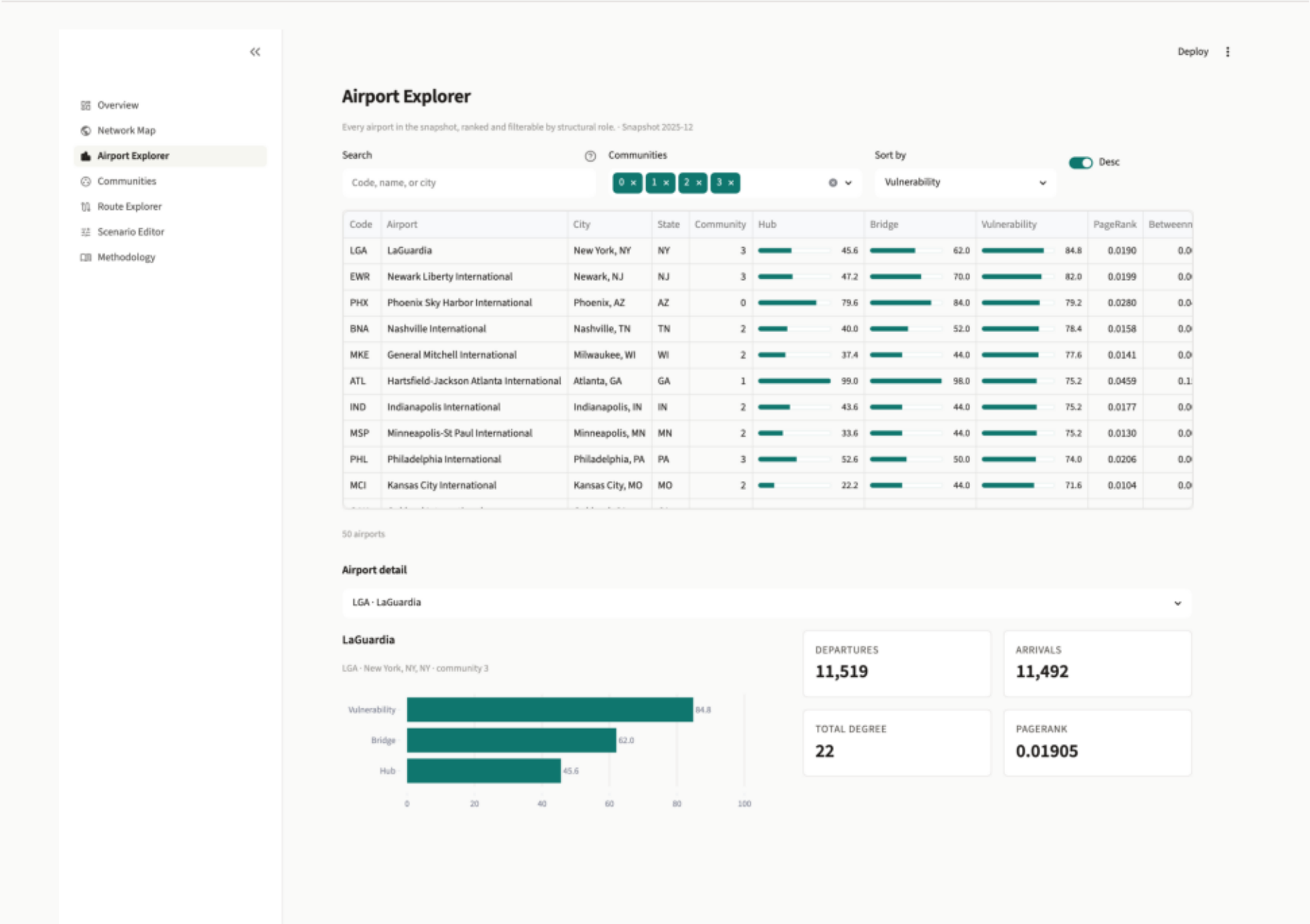
Network Map

The route network over an Albers USA projection. Marker area follows hub score, color follows Leiden community, and cross-community routes are tinted separately. The regional structure is legible directly from the map.



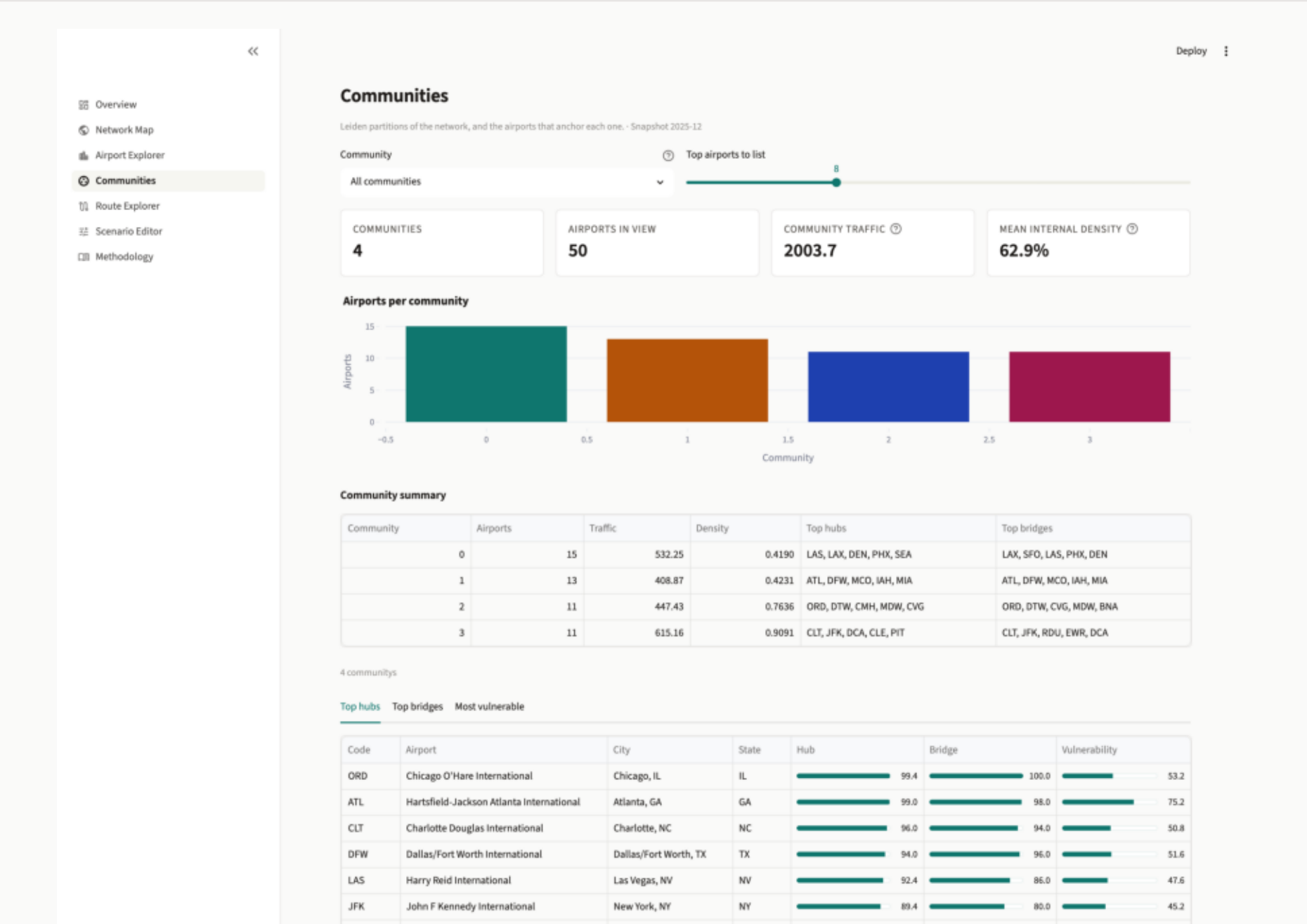
Airport Explorer

Every airport in the snapshot, searchable by code, name, or city. Percentile scores render as bars on a fixed 0-100 domain so standing is comparable between tables.



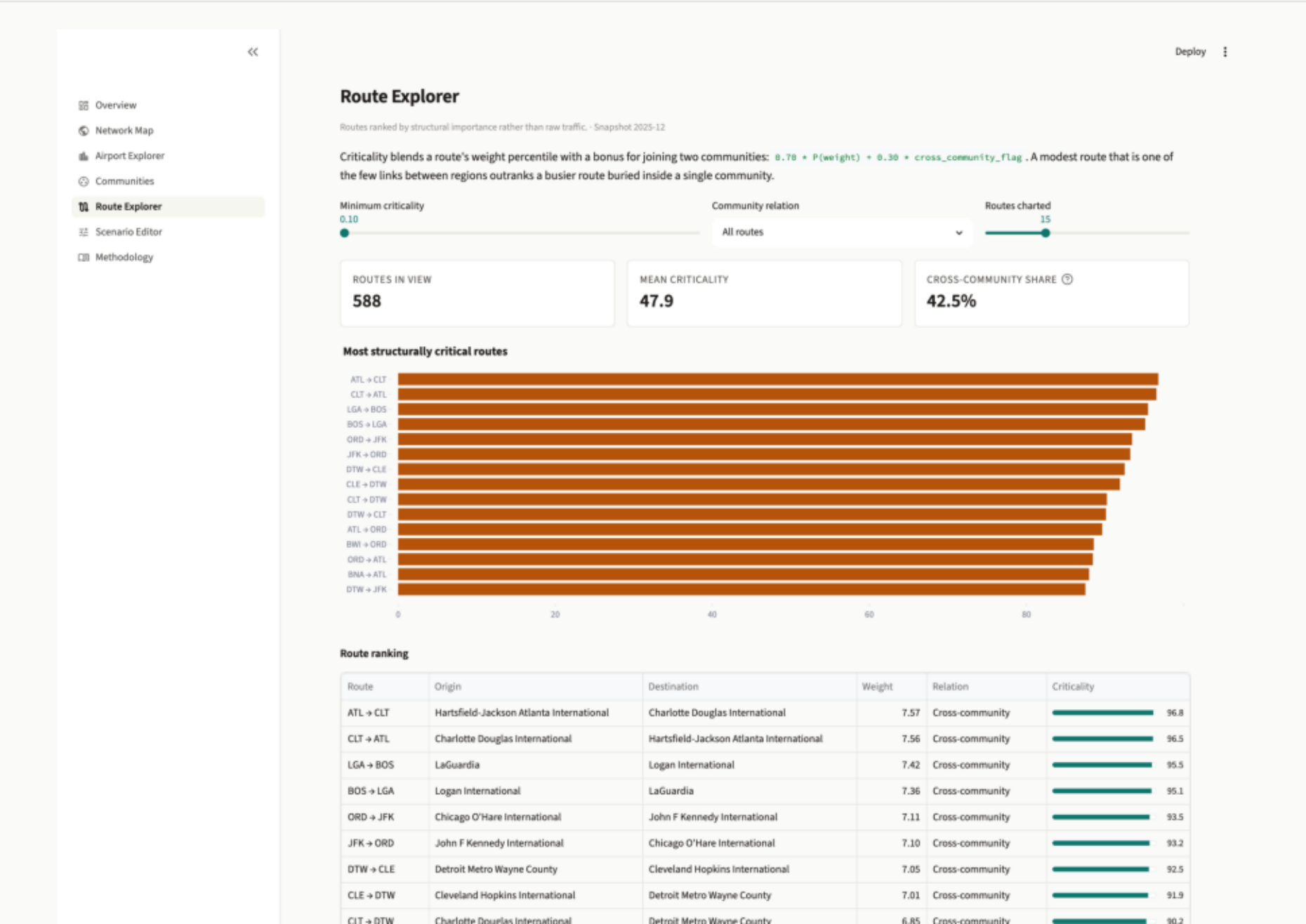
Communities

Leiden partitions with their size, internal traffic, and density. The top hub and bridge lists stored as airport ids in the artifact are resolved to codes for display.



Route Explorer

Routes ranked by structural importance rather than raw traffic. A modest route that is one of few links between regions outranks a busier route buried inside one community.



Scenario Editor

The baseline network before any simulation. Any airport can be removed by clicking it, or through the quick-find control.

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Overview

Network Map

Airport Explorer

Communities

Route Explorer

Scenario Editor

Methodology

Deploy

Scenario Editor

Simulate removing an airport or route. Click the map, or use quick-find below. · Snapshot 2025-12

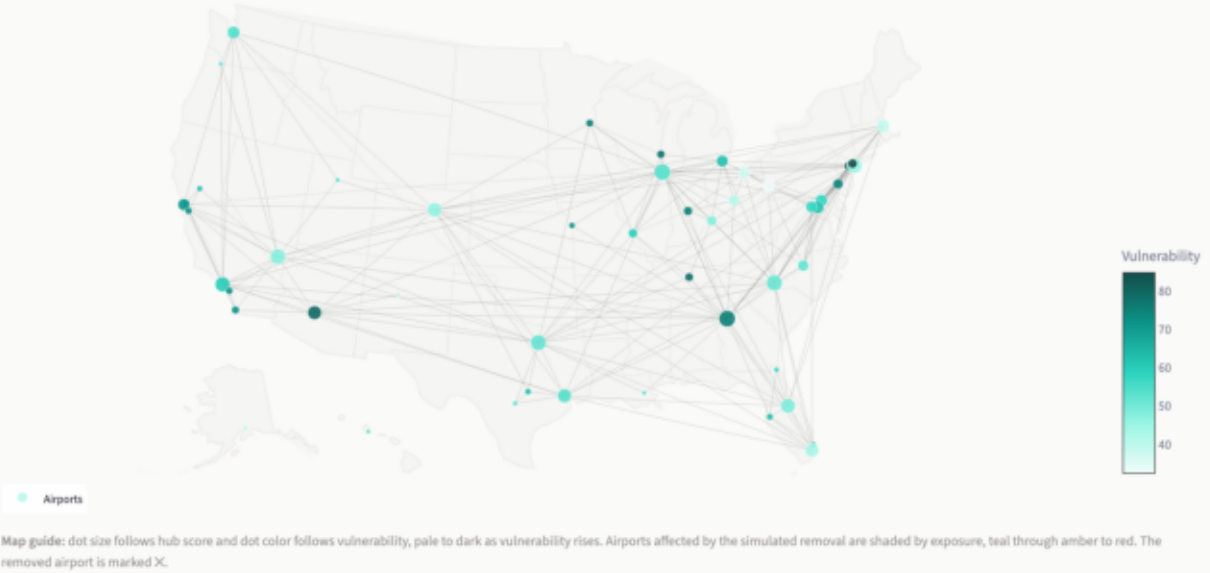
Quick-find airport

— search by name or code —

Show routes with analysis weight $\geq$

5.00

Click any airport to simulate its removal from the network



Airports

Map guide: dot size follows hub score and dot color follows vulnerability, pale to dark as vulnerability rises. Airports affected by the simulated removal are shaded by exposure, teal through amber to red. The removed airport is marked X.

No active simulation. Click an airport on the map or use quick-find above.

↶ Undo

Restore network

> Scenario history (0 runs)

Route removal

Remove a specific directed route and run the scenario engine.

Route to remove

ABQ → PHX

Methodology

Every formula as implemented, followed by the model's known limitations stated plainly rather than omitted.

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Methodology

Data/model assumptions, formulas, and caveats used by this implementation.

Data pipeline and graph model

- **Snapshot policy:** Monthly snapshots (`snapshot_id = YYYY-MM`) from processed artifacts.
- **Primary graph:** Directed weighted airport network where edge weight is `analysis_weight`.
- **Weight definition:** `analysis_weight = log1p(flight_count)`.
- **App behavior:** Pages read precomputed artifacts (`metrics.csv`, `communities.csv`, `route_metrics.csv`, `scenarios.csv`, `scenario_exposure.csv`) through app loaders with required-column guards.

Airport-level metrics

The metrics engine computes centrality and percentile-scaled composites for comparison:

$$P(\text{metric}) = \text{percentile rank scaled to } [0, 100]$$
$$\text{HubScore}(i) = 0.50 \cdot P(s_{\text{total}}(i)) + 0.30 \cdot P(\text{PageRank}(i)) + 0.20 \cdot P(deg_{\text{total}}(i))$$
$$\text{BridgeScore}(i) = P(\text{Betweenness}(i))$$
$$\text{Vulnerability}(i) = 0.60 \cdot P(\text{ImpactScore}(\text{remove } i)) + 0.40 \cdot P(\text{BridgeScore}(i))$$

Community and route metrics

Leiden communities partition airports into groups used by APP-04 and APP-05.

$$\text{CommunityTraffic}(C) = \sum_{(i,j) \in C} w(i,j)$$
$$\text{InternalDensity}(C) = \frac{\text{internal edges}(C)}{|C| \cdot (|C| - 1)}$$
$$\text{RouteCriticality}(i,j) = 0.70 \cdot P(w(i,j)) + 0.30 \cdot \text{CrossCommunity}(i,j)$$

`CrossCommunity(i,j)` is implemented as:

- `100` when origin and destination belong to different Leiden communities

Scenario Editor — active simulation

ATL removed. The removed airport is marked, airports touched by the two-hop ripple are shaded by exposure, and the impact and network-health scores update against the live graph.

